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Finetuned Multimodal Language Models Are High-Quality Image-Text Data Filters

Weizhi Wang[1] **Khalil Mrini**[2] **Linjie Yang**[2] **Sateesh Kumar**[2]
Yu Tian[2] **Xifeng Yan**[1] **Heng Wang**[2]
[1]University of California, Santa Barbara [2]Bytedance, US

https://mlm-filter.github.io

Abstract

We propose a novel framework for filtering image-text data by leveraging fine-tuned Multimodal Language Models (MLMs). Our approach outperforms predominant filtering methods (*e.g.,* CLIPScore) via integrating the recent advances in MLMs. We design four distinct yet complementary metrics to holistically measure the quality of image-text data. A new pipeline is established to construct high-quality instruction data for fine-tuning MLMs as data filters. Comparing with CLIPScore, our MLM filters produce more precise and comprehensive scores that directly improve the quality of filtered data and boost the performance of pre-trained models. We achieve significant improvements over CLIPScore on popular foundation models (*i.e.,* CLIP and BLIP2) and various downstream tasks. Our MLM filter can generalize to different models and tasks, and be used as a drop-in replacement for CLIPScore. An additional ablation study is provided to verify our design choices for the MLM filter.

1 Introduction

Large-scale image-text datasets [SDGS18a, SDGS18b, SVB+21, SBV+22, BPK+22] have been the major driving force for the recent breakthrough in Vision-Language Models (VLMs) and Text-to-Image generation models. The ever-growing size of such datasets allows researchers to scale the models to unprecedented capacities with billions or even trillions of parameters. These humongous foundation models lead to significant improvements in many down-stream tasks, such as image classification, text-to-image retrieval, image captioning, visual question answering, image generation and editing, *etc.* One great example is the OpenAI CLIP [RKH+21] model, which is trained with 400M web-crawled image-text pairs. The CLIP model demonstrates impressive zero-shot learning capability across a wide range of different tasks.

The quality of image-text data plays a decisive role in the final performance of foundation models. But web-crawled image-text data are often very noisy, *e.g.,* the corresponding text data is low quality or does not match the content of the image. How to build high-quality image-text datasets is a challenging research problem that attracts lots of interests recently. [XXT+23] try to re-create the data curation process from CLIP. [NIW+22] advocate that data quality is more important than quantity for model robustness. The DataComp challenge [GIF+23] is introduced to systematically evaluate different data-filtering techniques.

Each successful foundation model have their own secret recipes for data filtering. Before the invention of CLIP, most techniques are hand-designed or rule-based. For example, CC3M and CC12M design a series of heuristics for image-based, text-based and image&text-based filtering. Model-based filtering becomes popular since the introduction of CLIPScore [HHF+21], which leverages the CLIP model to compute the cosine similarity between image and text to measure their alignment.

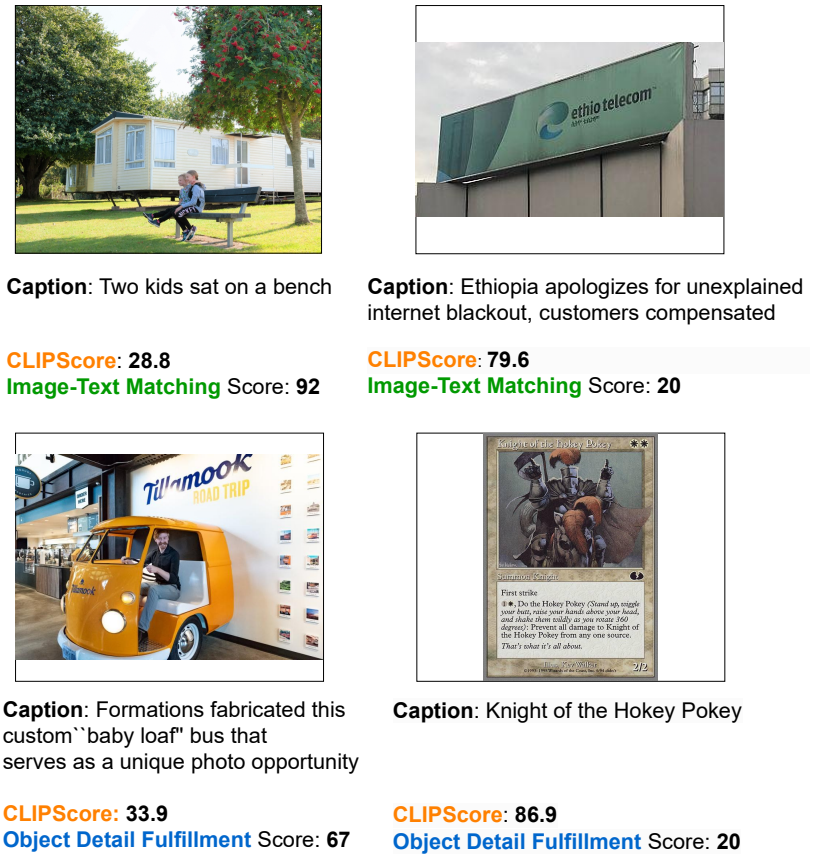


Figure 1: CLIPScore fails in differentiating the fine-grained object-level image-text alignment, while the image-text matching score generated by MLM Filter significantly captures such alignment.

CLIPScore has become the predominant method for filtering image-text data. However, recent research [TJS23, TLZ+24] finds that visual features from CLIP are blind to subtle differences in the image, *e.g.,* object number, shape and position. Because the contrastive loss is applied to the whole image, CLIPScore is less sensitive to capture the fine-grained object-level alignment information, shown in Figure 1. Additionally, the text encoder of CLIP can only process up to 77 tokens. The information loss from the text encoder can limit CLIPScore to process data with long captions. This limitation can be serious for Text-to-Image generation models [BGJ+23] that rely on long and highly-descriptive captions.

Compared with the contrastively trained CLIP model, Multimodal Language Models (MLMs) have demonstrated promising capability in predicting the quality of generated images or text and aligning well with human preferences. More specifically, the image-text matching scores generated by GPT-4Vision [Ope23] are more consistent with human experts compared with CLIPScore in recent MLM-based evaluation [YLL+23, ZLW+23]. This motivates us to integrate recent advances in MLMs for high-quality data filtering:

> *"Can we adapt strong MLMs to generate scores for assessing image-text data quality and outperform CLIPScore for image-text data filtering?"*

Though GPT-4V is better at measuring image-text alignment, directly applying GPT-4V-scale MLMs in filtering billions of image-text data is computationally too costly. A good filtering method should be both effective and efficient due to the sheer amount of data we need to process. There are smaller MLMs (*e.g.,* LLaVA [LLWL23], MiniGPT-4 [ZCS+23], etc), which are more efficient but fail to generate scores at a granularity that can reflect the subtle changes in the image-text data, since they are mainly instruction-tuned on task completion data. In this paper, we propose to combine the best of both worlds, leveraging proprietary LLMs or MLMs to construct high-quality instruction tuning

data for effectiveness, and fine-tuning more accessible open-source MLMs to inject the knowledge from the high-quality data for efficiency.

We summarize our major contributions as follows:

- We propose the MLM filter which incorporates the recent progress from MLMs for image-text data filtering and can be used as a drop-in replacement to the popular CLIPScore.
- We design four diverse metrics to measure the image-text data quality from different perspectives, and a new pipeline to construct high-quality instruction data to harvest the information from proprietary models.
- Foundation models trained with our MLM filtered data demonstrate significant improvements, *e.g.,* 1.7% better on 38 downstream tasks from DATACOMP comparing with CLIPScore.

2 Related Work

Data Filters. Initial work, such as ImageNet [DDS+09], relies on manual data filtering to select high-quality images and captions. More recent work [RKH+21, JYX+21] pushes the size of image-text dataset to the order of hundreds of millions, and thus employs fixed rules and heuristics for filtering. LAION [SVB+21] introduce the CLIPScore metric computed by the pre-trained CLIP model in filtering high-quality image-text pairs. CLIPScore filtering then becomes a widespread method of constructing large-scale web-crawled datasets [BPK+22, SBV+22, GIF+23]. Based on that, DATACOMP [GIF+23] is the first work to propose a benchmark for evaluating data filtering methods. [YTK+23] introduce a set of tools to improve data filtering including CLIP-FLIP, distribution matching, de-duplication and clustering. Similarly, [MGL+23] propose text masking to improve filtering. On the other hand, [FJJ+23] use high quality image-text pairs to train a new CLIP filtering network instead of using OpenAI's original CLIPScore. These papers all build upon CLIP filtering and introduce various techniques to improve it. In contrast, we investigate an alternate approach to CLIP-based Filtering, which employs fine-tuned Multimodal Language Models for large-scale image-text data filtering. Additionally, various works [CLY+23, WJHS23] deploys proprietary LLMs like GPT-4 to score and filter text-only and visual instruction data.

Multimodal Language Models. Recent Multimodal Language Models [ADL+22, HDW+24, WDC+22, LLSH23, ZCS+23, LLWL23] concatenate vision encoders with the latest LLMs via cross-model adapters to enable LLMs [TGZ+23, CLL+23, TMS+23] to take visual inputs. The most typical vision encoders deployed in MLMs are still the vision transformer models in CLIP pre-trained models [RKH+21] for extracting visual features of input images. Moreover, various adapter architectures are proposed to connect the feature space of different modalities, including Q-former proposed by BLIP-2 [LLSH23], a simple MLP layer used in LLaVA [LLWL23], and Visual Experts of CogVLM [WLY+23].

Multimodal Instruction Tuning. Instruction tuning [MKBH21, WBZ+21, OWJ+22] is a fine-tuning paradigm that enables LLMs to perform unseen tasks. This zero-shot performance is enabled by training LLMs using natural language instructions to explain the goal of the task. Instruction tuning is much more computationally efficient than full-set fine-tuning, and can enable LLMs to achieve zero-shot performance scores that are competitive with fully supervised models. LLaVA [LLWL23] introduces multimodal instruction tuning via fine-tuning MLMs on a set of visual instructions. MLMs that use instruction tuning [DLL+23, LLLL23] achieve SOTA performance on various vision-language tasks, such as visual question answering and visual reasoning.

3 Fine-Tuned Multimodal Language Models as Data Filters

3.1 Overview

We propose to adopt fine-tuned Multimodal Language Model as effective data filters to select high-quality image-text data to promote the VLM pre-training, which involves three stages: 1) constructing multimodal instruction tuning data on proposed quality scoring tasks to fine-tune MLM to realize accurate quality assessment; 2) adopt the fine-tuned MLM Filter to generate quality scores for each data point in the data pool and then select the high-quality data; 3) pre-train VLMs using the filtered

dataset and evaluate the pre-trained VLMs on downstream tasks to demonstrate the effectiveness of the proposed filtering method. The detailed pipeline for the three stages is shown in Figure 2.

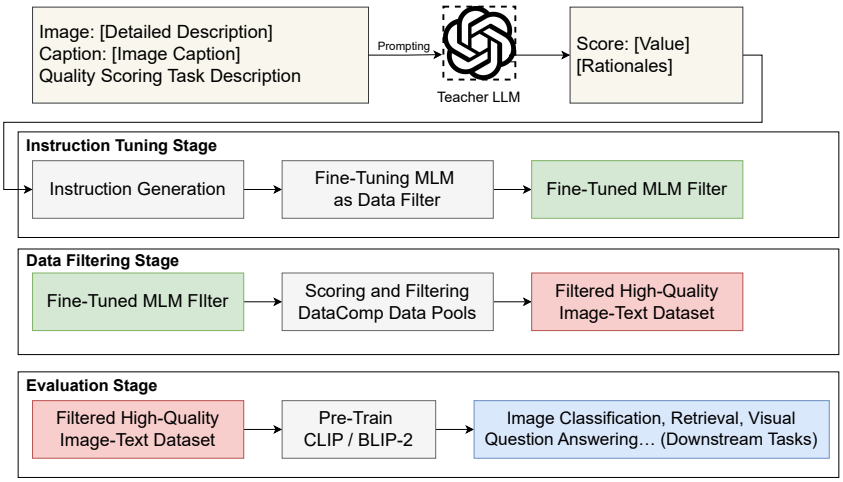


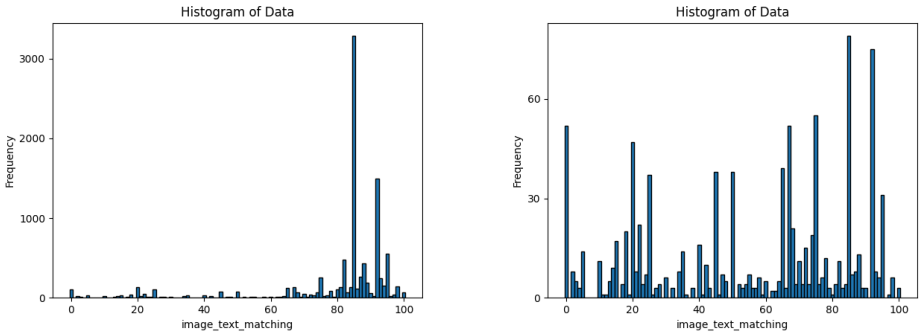
Figure 2: Illustration of the pipeline of fine-tuning MLM Filter and employing it for data filtering.

3.2 Constructing Multimodal Instruction Tuning Data for Scoring Tasks

In order to work as an effective data filter, the MLM must generate quality scores for every single image-text pair for data selection and filtering. To enable MLMs like LLaVA to reason accurately on the quality score, we propose to fine-tune such MLMs on a set of scoring tasks to enhance their scoring capability. The multimodal instruction tuning data needed for scoring tasks are hard and expensive to collect via human labeling, and thus we leverage proprietary models GPT-4 or GPT-4V to construct such multimodal instruction data for scoring tasks.

Defining Metrics for Image-Text Quality Assessment. Conventional data filters like CLIPScore focus on the overall holistic matching of image and text via computing the cosine similarity between hidden features of image and text. However, such implicit scoring is poor in discriminating hard or ambiguous samples, leading to the false negative score predictions shown in Figure 1. We propose to leverage strong Multimodal Language Models to predict the quality scores towards image-text pairs. Beyond the overall image-text alignment assessment, the fine-tuned MLM filters can evaluate the quality of image-text pairs from multiple perspectives. We propose four quality evaluation metrics to comprehensively evaluate the data quality:

- Image-Text Matching (ITM): the ITM metric focuses on evaluating whether the image caption accurately represents the main features and objects of the image and captures its primary theme. The fine-tuned MLM data filter can explicitly generate the ITM score on a scale of 100.
- Object Detail Fulfillment (ODF): the ODF metric focuses on evaluating whether the image caption provides detailed descriptions of objects that align with the image. Specifically, ODF assesses if the caption sufficiently describes the properties of the objects in the image, *e.g.,* number, color, size, position, shape, etc. Compared with the ITM metric, the ODF metric focuses more on the fine-grained alignment between the detailed object properties in the image and the ones described in the corresponding caption.
- Caption Text Quality (CTQ): the CTQ metric focuses on evaluating the text quality of image caption based on the grammatical correctness, diversity of vocabulary (*e.g.,* the range and uniqueness of words), fluency (*e.g.,* smoothness and natural flow of sentences), readability, length, and structure. Previous data-centric research [YTK+23] finds that web-crawled data is poor in its text quality, as it contains various bad text patterns, such as repeated words or textual noise. Thus, we propose to fine-tune MLMs to assess the text quality of image captions for data filtering.
- Semantic Understanding (SU): the SU metric focuses on determining if the image caption provides additional semantic information that is not readily apparent just from the image itself. Such auxiliary semantic information can be 1) the professions of persons in the image; 2) the locations,



(a) (b)

Figure 3: (a) image text matching score distribution of initial 10k instructions using GPT-4V on CC12M; (b) image text matching score distribution of final 1k instructions uniformly sampled from 10 buckets.

addresses, festivals, country names, city names; 3) the names or entities of buildings, people, bird species, animal breeds, car models, engines in the image; 4) the social relationships between the people in the image, *i.e.,* lovers, parent, or child. We suggest that adopting SU metric for data filtering can select image-text pairs with auxiliary semantics, which can further enhance the commonsense reasoning capability of pre-trained VLMs.

Prompting the Teacher Models. We select two state-of-the-art teacher models, GPT-4 and GPT-4V, to construct the multimodal instruction data for quality scoring tasks. Constructing multimodal instruction data with GPT-4V is much easier as GPT-4V can directly take visual inputs. As GPT-4 is a text-only LLM, we transform the image into a detailed text description to prompt a text-only GPT-4. The prompt for such dense captioning process is *Please generate a dense caption in 4-6 sentences for describing the image in detail as much as you can*. These comprehensive image descriptions are generated using a SOTA image captioning models, such as LLaVA or ShareGPT4V [CLD+23]. With the prompt to the teacher model and the generated output, the visual instruction data can be simply formatted as *User: {Prompt} Assistant: {Output}*.

Prompting Strategies. As the scoring tasks involve a reasoning process to predict final accurate quality metrics for an image-text pair, we consider two prompting strategies to ensure the reasoning accuracy of the fine-tuned multimodal language model: Chain-of-Thought (CoT) Reasoning [WWS+22], and Rationalization Reasoning [CRLB18]. The major difference between the two prompting strategies are the generation order of the score and the generated reasoning steps. The exemplar prompts for two prompting strategies are presented in Appendix B Table 7. Between these two prompting strategies, we select the rationalization reasoning as we find it to be the most efficient and accurate. Computational efficiency is a concern as the scoring MLM should be able to score billions of image-text pairs. If the MLM is fine-tuned to output the score value first, the model's text generation process can be stopped early in the inference stage as only the score value is needed for filtering. Secondly, the experimental results of LLaVA demonstrate that the instruction tuning with rationalization reasoning leads to better performance on the ScienceQA benchmark [SGM+22] than CoT reasoning. Four final prompts for different scoring metrics are presented in Appendix A.

Selecting Image-Text Pairs for Data Collection. The multimodal instruction data used for fine-tuning should contain image-text pairs of varying quality. Thus, data diversity is essential to enhance the fine-tuned MLM filter, enabling it to effectively score image-text data across all quality levels. We select two different image-text dataset as the data pool for constructing instruction tuning data: the Conceptual Captions 12M (CC12m) [SDGS18b], and the DATACOMP Medium 128M Dataset [GIF+23]. To enhance the diversity of the instruction set, we perform clustering and uniform-sampling on the sentence embeddings of each captioning text. The sentence embedding model we use is the pre-trained MPNet [STQ+20] encoder model, which is contrastively pre-trained on a mixture of retrieval and natural language inference datasets. We directly use the pre-trained MPNet provided by `Sentence Transformers` [RG20] to generate the sentence embedding towards each image caption. We set the number of clusters as $10k$ and $20k$ for CC12M and Datacomp-Medium, respectively. The

Captioner	Data Resource	#Sampling Buckets	Teacher Model	ImageNet-1k	ImageNet dist. shifts	VTAB	Retrieval	Average over 38 datasets
LLaVA	CC12M	10	GPT-4	29.0	24.5	35.0	29.3	34.2
ShareGPT4V	CC12M	10	GPT-4	28.4	24.9	35.3	28.2	33.7
N/A	**DataComp**	10	GPT-4V	29.6	24.8	34.2	26.7	33.2
N/A	**CC12M**	10	GPT-4V	30.5	25.3	33.4	28.0	33.7
ShareGPT4V	CC12M	**10**	GPT-4	28.4	24.9	35.3	28.2	33.7
ShareGPT4V	CC12M	**100**	GPT-4	27.5	23.0	34.6	28.8	33.2
LLaVA	CC12M	10	**GPT-4**	29.0	24.5	35.0	29.3	34.2
N/A	CC12M	10	**GPT-4V**	30.5	25.3	33.4	28.0	33.7

Table 1: Ablations on different design choices for constructing multimodal instruction data for quality scoring tasks.

image-text pairs for constructing instruction tuning data are uniformly sampled from each cluster, in which only one data point closest to the cluster centroid is selected.

Sampling Final Instructions for Scoring Tasks. As we find that the initial $10k$ instruction data generated by teacher models are not uniformly distributed on the score scale of 100 in Figure 3(a), we need to sample the initial instruction data into a balanced instruction set to avoid learning bias. Considering that the ideal size of multi-task instruction tuning dataset is $50k$ instructions [CLL+23, TMS+23], we decide to sample $1k$ instructions from $10k$ initial generated instruction data for each scoring tasks, which ensure the generalization capability of instruction-tuned MLM. Thus, there are $4k$ instruction data of quality scoring tasks to be included in the total $50k$ instruction dataset, such that there is 1k instruction data for each proposed quality metric. We experiment with two sampling methods to ensure that the instruction data distribution is balanced on the scoring scale of 100: 1) grouping all data into 10 buckets and uniformly sampling 100 instructions from each bucket; 2) grouping all data into 100 buckets and uniformly sampling 10 instructions from each bucket. The score distribution of sampled 10k instruction in Figure 3(b) are more diverse and uniform than the original score distribution in Figure 3(a). The code for sampling the final $4k$ instruction is presented in Appendix C.

Mixture with instruction data of multi-tasks. The multimodal instruction tuning process should involve a diverse set of tasks [DLL+23, LLLL23] to enhance the zero-shot reasoning capability of fine-tuned MLMs. In addition to 4k multimodal instruction data of the proposed data quality scoring tasks, we sample another 46k multimodal instructions from LLaVA-665k instruction datasets. We allocate a larger portion of our data mixture to reasoning tasks, such as complex reasoning [LLWL23] and GQA [HM19] as we regard that enhancing reasoning capabilities will improve the scoring capability of our fine-tuned MLM. The detailed statistics on the size of each dataset sampled for data mixture are presented in Appendix D Table 8.

3.3 Instruction-Tuning on Multimodal Language Models

We adopt LLaVA-1.5 based on Vicuna-13B LLM [CLL+23, LLLL23] as the Multimodal Language Model architecture for instruction tuning on the mixed instructions of data quality scoring tasks and other multimodal tasks. The training process of LLaVA-1.5 involves pre-training on image-text pairs and instruction tuning on multimodal instructions. We directly take the pre-trained checkpoint and only reimplement the instruction tuning stage with our mixed instruction set.

3.4 Creating Optimal MLM Data Filters

We propose various different design choices for constructing instruction data for data quality scoring tasks in Section 3.2. These design choices may make a significant difference in the effectiveness of instruction tuning. To create the optimal fine-tuned MLM data filter, we conduct comprehensive ablation studies to investigate the effects of different design choices on the filtering performance. Four major design choices for constructing the instruction data for scoring tasks are investigated: 1) we experiment with two captioning models to transform image into text-base detailed description for prompting GPT-4, including LLaVA and ShareGPT4V [CLD+23]; 2) we experiment with two different image-text datasets for constructing visual instructions, including CC12M and DataComp Medium 128M; 3) we experiment with two different numbers of grouping buckets, 10 and 100, for

Filter	Metrics	Teacher Model	ImageNet-1k	ImageNet dist. shifts	VTAB	Retrieval	Average over 38 datasets
No Filtering	-	-	17.6	15.2	25.9	21.9	25.8
Basic Filtering	Rules	-	22.6	19.3	28.4	25.1	28.5
LAION Filtering	CLIPScore+Rules	-	23.0	19.8	30.7	23.3	29.2
CLIPScore	CLIPScore	-	27.3	23.0	33.8	25.1	32.8
MLM-FILTER	Image-Text Matching	GPT-4	28.6	23.7	34.4	**30.0**	33.4
MLM-FILTER	Object Detail Fulfillment	GPT-4	29.0	24.5	35.0	29.3	34.2
MLM-FILTER	Caption Text Quality	GPT-4	25.2	20.9	32.1	26.4	30.9
MLM-FILTER	Semantic Understanding	GPT-4	20.3	16.1	28.4	20.2	27.0
MLM-FILTER	Image-Text Matching	GPT-4V	29.4	24.4	**36.1**	29.7	34.2
MLM-FILTER	Object Detail Fulfillment	GPT-4V	**30.5**	25.3	33.4	28.0	33.7
MLM-FILTER	Caption Text Quality	GPT-4V	24.3	20.4	32.3	24.5	30.9
MLM-FILTER	Semantic Understanding	GPT-4V	16.2	13.9	23.3	18.7	24.0
MLM-FILTER	ITM AND ODF	GPT-4V	30.3	**25.6**	36.0	29.0	**34.5**
MLM-FILTER	ITM OR ODF	GPT-4V	28.9	24.5	35.2	29.0	33.9

Table 2: Zero-shot performance of CLIP models pre-trained using baseline filtering methods and proposed MLM-FILTER on *Medium* scale pools of the DataComp benchmark. AND represents the combination of ITM and ODF metrics using AND operation.

sampling the final 4k instructions; 4) we experiment with different teacher models to get multimodal instructions, including GPT-4 and GPT-4 Vision. Additionally, we use the DataComp benchmark to evaluate the effectiveness of different data filtering hyperparameters.

DataComp Benchmark. The DataComp benchmark [GIF+23] has been introduced to systematically compare the performance of different data filtering methods. In this benchmark, the training code and computational budget is fixed across all competing methods to facilitate direct comparison between methods. The DataComp provides a fixed original image-text data pool for different filtering methods to ensure a fair comparison. The performance is measured by training a CLIP model on the filtered dataset and then testing the zero-shot capabilities of this CLIP model on a suite of 38 classification and retrieval tasks. We select the Medium scale training setting to train ViT-B/32 CLIP models on datasets resulting from various MLM data filter configurations.

Ablation Results. To investigate the effects of each design choice, we keep the selection of the other three design choices the same and only change one design choice for each experiment group. As we propose four different metrics to assess data quality, we only adopt the metric of *Object Detail Fulfillment* as the filtering metric to select a high-quality subset from the 128M medium scale data pool. The ablation results for all four design choices are presented in Table 1.

The first two lines in Table 1 demonstrate that adopting LLaVA as the captioning model to transform images into detailed descriptions for instruction data construction leads to better filtering performance. Next, adopting CC12M to sample image-text pairs for data construction outperforms the design choice of using DataComp-Medium dataset. We suppose it is because the image quality of CC12M is significantly better than that of DataComp, enabling the instruction tuning process more knowledge intensive. Thirdly, grouping the initial instructions into 10 buckets for sampling illustrates priority over using 100 buckets. In terms of the selection of teacher models, the MLM filters learned from different teacher models exhibit distinct strengths across different tasks. The MLM filter learned from GPT-4 performs better in VTAB [ZPK+19] classification and retrieval datasets, while the MLM filter learned from GPT-4V obtains higher scores in ImageNet [DDS+09] related datasets. Finally, we decide to fix the other three choices as LLaVA captioner, CC12M data resources, and 10 sampling buckets. We report the two versions of MLM-based filters with different teacher models GPT4 and GPT-4V for future experiments, denoted as MLM-FILTER-GPT4 and MLM-FILTER-GPT4V respectively.

4 Experiments

In this section, we evaluate the effectiveness of adopting fine-tuned MLMs as high-quality image-text data filters. We compare the performance of vision-language models pre-trained on datasets filtered using a baseline filter with their performance using our MLM filter. We select two different VLM

architectures for comprehensive evaluation: CLIP pre-training and BLIP-2 pre-training. Additionally, we conduct human evaluation to compute the correlation between the scoring generated by our proposed MLM filter model and the baseline CLIP model.

4.1 CLIP Pre-Training on DataComp Medium and Large Scales

Filter	Metrics	Teacher Model	ImageNet-1k	ImageNet dist. shifts	VTAB	Retrieval	Average over 38 datasets
No Filtering	-	-	45.9	37.8	42.6	41.9	43.7
Basic Filtering	Rules	-	51.6	42.3	44.6	48.0	45.8
LAION Filtering	CLIPScore+Rules	-	55.3	45.3	51.0	49.5	50.1
CLIPScore	CLIPScore	-	57.8	47.4	53.8	46.6	52.9
MLM-FILTER	Object Detail Fulfillment	GPT-4	58.9	48.9	57.4	52.5	54.2

Table 3: Zero-shot performance of CLIP models pre-trained using baseline filtering methods and proposed MLM-FILTER on *Large* scale pools of the DataComp benchmark.

Evaluation Setup. We select the DataComp benchmark to evaluate the effectiveness of adopting fine-tuned MLM as data filter. The evaluation process involves the data filtering stage and evaluation stage, which are shown in Figure 2. During the data filtering stage, we adopt the MLM-Filter to generate quality scores on all 128M medium-scale data and 1.28B large-scale data. After that, an integer filtering threshold is calculated based on the closest value that retains 30% of the overall data pool, 38.4M for Medium and 384M for Large. Such threshold is set up to select all the image-text pairs, of which the quality score is larger or equal to the threshold. We report the results using each defined metric to filter data separately and we consider two MLM filters learning from different teacher models. Additionally, we also report the results of experiments with a combination of two metrics for data filtering. Finally, we select a high-quality subset from the medium or large scale image-text data pools based on different proposed quality metrics. During the evaluation stage, we adopt the selected high-quality data subset to pre-train a CLIP model and compare the performance of our CLIP model with CLIP models pre-trained on datasets filtered by other methods.

Baselines. We compare the proposed MLM filter with other baseline filtering methods from DataComp, including applying no filtering, basic filtering, LAION filtering and CLIPScore filtering. The basic filtering method adopts three rule-based filters, filtering English only, filtering by caption length, and filtering by image size. The LAION filtering adopts both the CLIPScore filtering using ViT-B/32 CLIP model and the English filtering. The CLIPScore filtering utilizes a larger ViT-L/14 CLIP model for score generation and data filtering.

Training Details. We strictly follow the training setup provided by DataComp. The computational budget and hyperparameters are fixed for pre-training CLIP using different filters. The CLIP model architecture is determined by the data scale, in which the ViT-B/32 model is pre-trained on the medium scale setting and ViT-B/16 model is on the large scale setting. We use 32 Nvidia A100 GPUs to train our models.

Results on DataComp Medium and Large Scale. The DataComp results between the proposed MLM filter and other baselines are presented in Table 2 and Table 3 for Medium and Large scale respectively. On the medium-scale DataComp benchmark, the proposed MLM Filter significantly outperforms the CLIPScore baseline on different task subgroups, achieving notable improvements of +3.2 accuracy on ImageNet-1k, +2.6 average accuracy on 6 ImageNet shifted datasets, +2.3 average accuracy on 13 VTAB datasets, and +4.9 average scores on 3 retrieval datasets. Moreover, the proposed MLM FILTER surpasses CLIPScore baseline by +1.7 and +1.3 improvements on the average scores over 38 datasets on DataComp Medium and Large Scale benchmarks, which demonstrates the proposed MLM Filter can work as more effective filtering method than CLIPScore filter. Additionally, we can draw the following auxiliary conclusions from the results:

The MLM Filter learned from GPT-4V performs better on ImageNet related datasets than the MLM Filter learned from GPT-4. The MLM-FILTER-GPT4V achieves the best performance on both ImageNet-1k and 6 ImageNet Shifted datasets. Both filtering metrics of Image Text Matching and Object Detail Fulfillment generated by MLM-FILTER-GPT4V outperforms the best ImageNet-1k accuracy of MLM-FILTER-GPT4, achieving a notable improvement of +1.1 accuracy.

Filter	Metrics	SVHN	MNIST	Avg.
MLM-FILTER-GPT4	ITM	8.2	10.3	9.2
MLM-FILTER-GPT4	ODF	14.6	19.3	16.9
MLM-FILTER-GPT4V	ITM	15.4	8.3	11.8
MLM-FILTER-GPT4V	ODF	9.0	6.8	7.9
MLM-FILTER-GPT4V	AND	12.9	11.6	12.3

Table 4: Zero-shot performance of pre-trained CLIP on SVHN and MNIST digit classification datasets. Avg. represents the average performance on two digit datasets. AND represents the combination of ITM and ODF metrics using AND operation.

The optimal filtering metric varies for fine-tuned MLM Filter learned from different teacher models. For the proposed MLM FILTER learned from different teacher models, the optimal filtering metric under single metric filtering setting is different. The Image-Text Matching is the optimal filtering metric for MLM-FILTER-GPT4V, while the Object Detail Fulfillment metric helps the MLM-FILTER-GPT4 most. The other two metrics of Caption Text Quality and Semantic Understanding cannot work as effective filtering quality metrics in DataComp benchmark, leading to worse performance than CLIPScore baseline. We regard that it is because the most of DataComp evaluation datasets are image classification datasets, which did not aligh with the filtering directions and objectives of CTQ and SU metrics.

Image-Text Matching is the best filtering metric for retrieval tasks. Our proposed MLM Filter achieves the SOTA performance on the three image-to-text and text-to-image datasets under DataComp Medium setting. The two types of MLM Filters achieves 30.0 and 29.7 average performance on three retrieval tasks using the ITM filtering metric, surpassing the CLIPScore baseline by 4.9 average scores. We also observe in results of both MLM Filter variants that the image-text matching metric leads to better performance on retrieval tasks compared with other three filtering metrics.

Combing different quality metrics effectively filters and identifies image-text pairs of better quality. The AND operation to combine ITM and ODF quality metrics means that the ITM and ODF score of selected datapoints should exceed the filtering thresholds of both metrics, while the OR operation to combine two metrics means that the selected datapoints should either exceed the threshold for ITM metric or that for ODF metric. The combination of ITM and ODF metrics using AND operation outperforms all the baseline filtering methods and other variants of MLM Filters, achieving the best average performance of 34.5 over 38 datasets.

The worse performance on digit classification tasks prevents MLM-FILTER-GPT4V from remarkably outperforming MLM-FILTER-GPT4. Even if MLM-FILTER-GPT4V outperforms MLM-FILTER-GPT4 on 23 ImageNet, VTAB and retrieval datasets, it only achieves the same average performance over 38 datasets as MLM-FILTER-GPT4. It is because the performance of MLM-FILTER-GPT4V on the two digit classification datasets significantly lags behind MLM-FILTER-GPT4 by 5.1 average score, shown in Table 4, which leads to 0.27 average score behind on 38 datasets. The combination of two quality metrics promotes the digit classification performance of MLM-FILTER-GPT4V, but does not resolve it.

4.2 BLIP2 Pre-Training

To demonstrate the effectiveness of our proposed MLM Filter across various VLM model architectures, we pre-train BLIP-2 VLM on the filtered dataset and evaluate the zero-shot performance of such BLIP-2 model on VQA datasets to compare the effectiveness of filtering methods on high-level vision-language tasks.

Training setup. We directly use the filtered dataset from DataComp Large 1.28B data pool using CLIPScore filtering and our proposed MLM Filtering. The batch size and number of pre-training steps are kept as the same as original implementation [LLSH23] for both the CLIPScore filtered dataset and MLM filtered dataset, in which both BLIP-2 models are iterated on 420M images for pre-training stage 1 and 154M images for stage 2. We use the same hyperparameters and number of GPUs for training. The visual encoder and LLM we used for BLIP-2 architecture are Eva-CLIP

Filter	Metric	VQA	GQA
CLIPScore	CLIPScore	55.1	34.8
MLM-FILTER-GPT4	ODF	56.8	36.2

Table 5: Zero-shot VQA performance of BLIP-2 models pre-trained on dataset filtered by different filtering methods.

Filter	Metric	Pearson	Spearman
CLIPScore	-	0.164	0.072
MLM-FILTER-GPT4	ITM	**0.452***	**0.430***
MLM-FILTER-GPT4	ODF	0.410*	0.384*
MLM-FILTER-GPT4V	ITM	0.328*	0.331*
MLM-FILTER-GPT4V	ODF	0.368*	0.374*

Table 6: Pearson and Spearman correlations between human-labeled quality scores and scores generated by MLM-Filter and CLIP. Images are scored on a scale of 100 for our MLMFilter, while CLIPScore is also normalized to the scale of 100. The * denotes significant correlations at $p < 0.05$.

ViT-g/14 [SFW+23] and Vicuna-7b [CLL+23] respectively. More training details are available in Appendix E Table 9.

Results. Two BLIP-2 models pre-trained on different filtered datasets are evaluated on VQAv2 [GKSS+17] and GQA [HM19] datasets in zero-shot manner and the results of zero-shot VQA performance are shown in Table 5. The BLIP-2 pre-trained with MLM-FILTER-GPT4 filtered image-text data achieves +1.7 and + 1.4 improvements on VQAv2 and GQA datasets than the BLIP-2 pre-trained on CLIPSCore filtered dataset.

4.3 Correlation with Human Scoring

We follow [ZLW+23] to compute the correlation between human scoring and model scoring to evaluate the alignment between human and the filtering model. A set of 100 image-text pairs are sampled from CC12M and MSCOCO [LMB+14] and labeled with human scores in terms of the image-text matching. CLIPScore and fine-tuned MLM filters are used to generate the image-text matching scores for the selected image-text pairs. Then, the Pearson and Spearman scores are reported between the human scores and model scores, as presented in Table 6. Our proposed MLM-FILTER scores are significantly aligned and correlated with human quality scores, while CLIPScore does not demonstrate such correlations. The two quality metrics Image-Text Matching and Object Detail Fulfillment all demonstrate significant correlations in similar levels.

4.4 Analysis

Effects of filtering fraction. We perform an ablation study to investigate the effects of the fraction of samples selected for pre-training CLIP on DataComp Medium benchmark performance. We select five fractions $\{0.2, 0.25, 0.3, 0.35, 0.4\}$ of the total 128M images of DataComp medium pool. The results are presented in Table 4. The top-30% of images selected for CLIP training achieve the best performance, which is also observed in [GIF+23]. Even adding 5% poison data leads to a huge performance drop on both ImageNet and average over 38 datasets.

Efficiency of MLM Filters. The MLM Filter used for quality score generation is LLaVA-1.5 with 14B model parameters , while CLIPScore adopts a CLIP ViT-L/14 model with 492M parameter in total. Even if the model size of the proposed MLM Filter is much larger than that of CLIPScore, due to the computation redundancy of the CLIP's dual-encoder architecture, the timecost for generating scores for 10k image-text pairs is average 24.3 mins for MLM Filter versus 11.2 mins for CLIPScore-ViT/L using one A100 GPU. Additionally, with the help of the latest techniques in language model inference acceleration, the TensorRT-LLM toolkit[1], we accelerate the score generation of our MLM

[1]https://github.com/NVIDIA/TensorRT-LLM

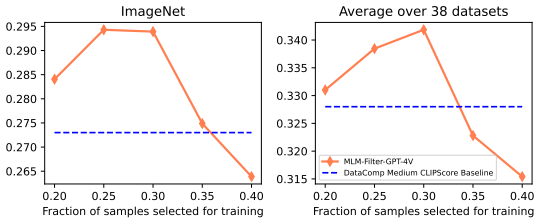


Figure 4: Effects of fraction of images selected for training CLIP.

Filter 4 times over, resulting in 6.1 mins in average for 10k samples. Thus, the proposed MLM Filter can achieve much better efficiency than CLIPScore.

5 Conclusion

We propose to instruction-tune Multimodal Language Model on quality scoring tasks and further leverage these fine-tuned MLM as effective data filters to select high-quality image-text pairs from large-scale web-crawled dataset. We find that, on CLIP and BLIP-2 models, pre-training on datasets filtered by our proposed MLM Filter significantly outperforms pre-training on CLIPScore-filtered datasets, demonstrating the superiority of our proposed MLM Filter over CLIPScore filtering.

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A Prompt Construction

After manually writing the first version of prompts, we leverage the GPT-4 to refine the human-written prompts. The final prompts for four quality scoring tasks are shown below:

Image Text Matching

Please evaluate if the provided text caption accurately represents the main features and objects of the image. The caption doesn't need to detail every aspect of the image, but it should capture its primary theme. Rate the overall quality of the text caption's match to the image on a scale of 1-100, considering the criteria mentioned.

Object Detail Fulfillment

Please evaluate the text caption to determine if it provides detailed descriptions of objects that align with the image description. Specifically, assess if the caption sufficiently describes the color, size, position, shape, material, etc., of the objects. Afterward, rate the caption's overall accuracy in capturing object details from the image on a scale of 1-100, based on the criteria provided.

Caption Text Quality

Please evaluate the text caption based on the following criteria: Grammatical Correctness, Diversity of Vocabulary (e.g., the range and uniqueness of words used), Fluency (e.g., smoothness and natural flow of sentences), Readability, Length, and Structure. Assign an overall quality score on a scale of 1-100.

Semantic Understanding

Please evaluate the given text caption in relation to its corresponding image description. Your goal is to determine if the text caption provides additional semantic information that isn't readily apparent just from the image itself.
For example:
1. If the image description mentions "a man" but the caption elaborates he is a "homeless man" or a "businessman," then the caption is enriching the semantic context.
2. If the caption introduces concepts like the mathematical tangent function, which require in-depth knowledge to deduce, it is imparting external semantics.
3. Captions revealing specific location addresses, festival details, or other nuanced data not easy to infer from the image also provide external semantic information.
4. Directly identifying specific entities in the image such as buildings, people, bird species, animal breeds, car models, engines, etc., in the caption introduces additional insights.
5. Should the image act as a contextual backdrop and the caption describes elements not explicitly showcased in the image, it has semantic depth.
6. Lastly, if the caption depicts relationships between the subjects in the image, which need commonsense knowledge to understand, it should be considered semantically rich.
Please assess and determine the extent of semantic enrichment the caption provides over the image description. Rate the text caption's semantic depth on a scale from 1 to 100.

B Examples for Two Prompting Strategies

Example for **Chain-of-Thought Reasoning**	Example for **Rationalization**
Please evaluate if the provided text caption accurately represents the main features and objects of the image. The caption doesn't need to detail every aspect of the image, but it should capture its primary theme. Rate the overall quality of the text caption's match to the image on a scale of 1-100, considering the criteria mentioned. **Please think step by step to first output your reasons to give such a score. In the subsequent line, please output a single line containing the value indicating the scores.**	Please evaluate if the provided text caption accurately represents the main features and objects of the image. The caption doesn't need to detail every aspect of the image, but it should capture its primary theme. Rate the overall quality of the text caption's match to the image on a scale of 1-100, considering the criteria mentioned. **Please first output a single line containing the value indicating the scores. In the subsequent line, please provide a comprehensive explanation of your evaluation, avoiding any potential bias.**

Table 7: Prompts for zero-shot Chain-of-Thought reasoning and Rationalization reasoning for assessing the image-text matching score.

C Sampling Final Instructions for Scoring Tasks

```
final_data = []
buckets = 10 * [[]]
for d in data:
    buckets[d['score']//10].append(d)
threshold = 130
num_downsample_bucket = 0
for b in buckets:
    if len(b) < threshold:
        final_data += p
    else:
        num_downsample_bucket += 1
downsampling_num = 1000 - len(final_data)
num_per_bucket = downsampling_num // num_downsample_bucket
for b in buckets:
    if len(b) > threshold:
        final_data += random.sample(b, num_per_bucket)
total_data += final_data[:1000]
```

D Data Mixture of Multi-Task Multimodal Instructions

E BLIP-2 Training Details

The detailed training hypperparamters and settings of BLIP-2 stage 1 and stage 2 are presented in Table 9.

Data	Size	Task
Visual Conversation [LLWL23]	5K	Conversation
Complex Reasoning [LLWL23]	16k	Visual Reasoning
Detail Description [LLWL23]	5k	Captioning
ShareGPT [Sha23]	10K	Language-Only Instructions
VQAv2 [GKSS+17]	2K	VQA
GQA [HM19]	3K	Visual Reasoning
OKVQA [MRFM19]	2K	Knowledge Grounded VQA
OCRVQA [MSSC19]	1K	OCR
TextCaption [SHRS20]	2K	Captioning
ITM Scoring	1k	Data Quality Scoring
ODF Scoring	1k	
CTQ Scoring	1k	
SU Scoring	1k	
Total	50k	

Table 8: Multimodal instruction data mixture of the data quality scoring tasks and other multimodal tasks.

Hyperparameter	BLIP-2
Stage-1 Pre-training	
# Trainable Parameters	188M
Precision	`float16`
Global Batch Size	1680
# Training Steps	250k
# GPUs	16
# Gradient Accumulation Steps	1
Min LR	1e-5
Peak LR	1e-4
# Warmup Steps	2000
LR Scheduler	Cosine LR Decay
Weight Decay	0.05
Adam (β_1, β_2)	(0.9, 0.98)
Stage-2 Pre-training	
# Trainable Parameters	188M
Precision	`float16`
Global Batch Size	1920
# Training Steps	80k
# GPUs	16
# Gradient Accumulation Steps	4
Min LR	5e-5
Peak LR	1e-4
# Warmup Steps	2000
LR Scheduler	Cosine LR Decay
Weight Decay	0.05
Adam (β_1, β_2)	(0.9, 0.98)

Table 9: Training details for BLIP-2 pre-training stage 1 and stage 2.